

Notes on the Value Function and Action-Value Function

Kieran Paranjpe

July 2026

1 Introduction

In reinforcement learning algorithms, we often use a 'critic' to improve quality. This can take the form of either a value function, or an action-value function. These are functions which quantify how good or bad it is to be in a state (or be in a state and take an action). These notes will go into further detail on these functions.

2 Background

2.1 The Markov Decision Process (MDP)

The reinforcement learning problem is the problem of finding the optimal **policy** to solve a given **Markov Decision Process** (MDP). An MDP can be viewed as a 4-tuple:

$$M = (S, A, \mathcal{P}(s'|a, s), \mathcal{R}(a, s, s'))$$

where S is the set of states and A is the set of actions [1, 2]. $\mathcal{P}(s'|a, s)$ is the probability of ending up in state s' given that we are in state s , and take action a . $\mathcal{R}(a, s, s') : (A \times S \times S) \rightarrow \mathbb{R}$ is the reward function that gives the reward for being in state s , taking action a , and ending up in state s' . We often use the shorthand $r_t = \mathcal{R}(a_t, s_t, s_{t+1})$ to denote the reward at timestep t . The policy is defined as the distribution over actions given the state [1], parameterised by weights θ :

$$\pi_\theta(a|s)$$

When we “run” the environment, we collect a sequence of states, actions and rewards: $s_1, a_1, r_1, \dots, s_T, a_T, r_T, s_F$, for T timesteps, where s_F is the final state. Each timestep, we sample $a_t \sim \pi_\theta(\cdot|s_t)$ to (hopefully) pick the action that maximises the total reward.

A note on notation: I will use $\mathcal{P}(x|y)$ to denote the probability of x given y , and $\mathcal{P}(\cdot|y)$ to denote the entire distribution from which x is sampled.

3 The Value Function

The value function quantifies how good it is to begin in a given state s , and then follow and sample actions from a policy π until the episode ends. It is defined as:

$$V^\pi(s) = \mathbb{E}_\pi\left[\sum_{k=t}^T \gamma^{k-t} r_k \mid S_t = s\right] = \mathbb{E}_\pi[G_t \mid S_t = s]$$

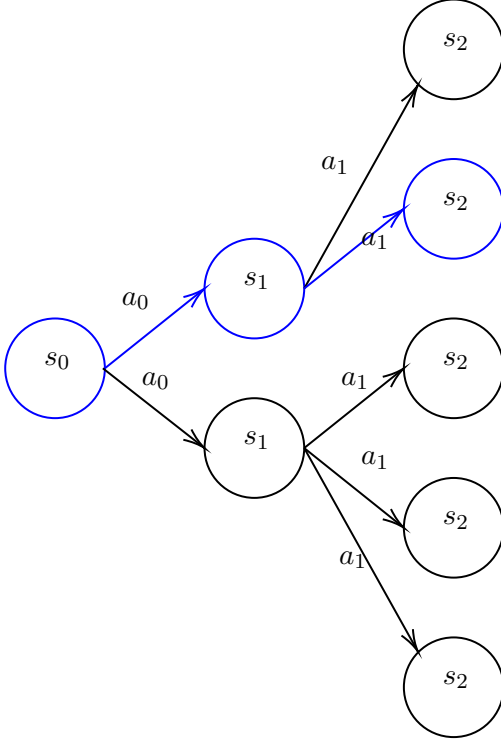
$$G_t = \sum_{k=t}^T \gamma^{k-t} r_k$$

where G_t is called the return, and T is the length of the episode (in principle this can be ∞) [1, 2]. The value function is the expected return when following the policy π , starting with $S_t = s$. We can empirically compute one specific value for G_t by doing a full episode rollout and computing the sum of rewards. The value γ , called the discount factor, is a hyperparameter that allows us to weigh earlier rewards more than later rewards.

In the definition of the value function, we must take the expectation over π , $\mathbb{E}_\pi$, but we don't explicitly define what this means. We claim it means to follow the policy, but what does that really mean?

3.1 Sample Rollout

First, let's take a look at a sample rollout:



In blue, we can see one specific rollout: s_0, a_0, s_1, a_1, s_2 . In black, we can see the other states and actions which the agent could have chosen. Each of these paths has its own return associated with it. The goal of the value function is to quantify how much return we expect to get if we consider the possibility of *going down all paths*, not just one.

3.2 Formal Value Function Expansion

To properly expand the value function, we must first define a new distribution:

$$\mathcal{L}(s \rightarrow s', k, \pi) = \mathbb{P}(s \rightarrow s', k, \pi)$$

This is the probability of starting in state s and ending up in state s' after k steps, while following the policy π . I use $\mathcal{L}$ to clearly distinguish from $\mathcal{P}$ from the MDP. If you notice, this distribution still relies on "following the policy π ", but it can be computed entirely from the MDP as follows:

$$\mathcal{L}(s \rightarrow s', k, \pi) = \sum_{a \in A} \pi(a|s) \sum_{s^* \in S} \mathcal{P}(s^*|s, a) \mathcal{L}(s^* \rightarrow s', k-1, \pi)$$

$$\mathcal{L}(s \rightarrow s', 0, \pi) = \begin{cases} 1 & \text{if } s = s' \\ 0 & \text{otherwise} \end{cases}$$

What is this really saying? $\sum_{a \in A} \pi(a|s) \sum_{s^* \in S} \mathcal{P}(s^*|s, a)$ means: start in state s , and then consider the probability of each possible action we could take, and then for each of those actions, consider the probability of ending up in each state. It enumerates over all the possible states we could end up in next, and sums the probability we end up there on the next step. Once we have this probability, we just weight the probability of going from this next possible state to our end state, in $k-1$ steps, so we multiply by $\mathcal{L}(s^* \rightarrow s', k-1, \pi)$. The base case is trivial: we can reach only the state we began with in 0 steps.

Now that we've established this distribution, we can fully expand the value function as follows:

$$\begin{aligned}
V^\pi(s) &= \mathbb{E}_\pi \left[\sum_{k=t}^T \gamma^{k-t} r_k \mid S_t = s \right] && \text{Original definition} \\
&= \sum_{k=t}^T \gamma^{k-t} \mathbb{E}_\pi [r_k \mid S_t = s] && \text{Pull out summation} \\
&= \sum_{k=t}^T \gamma^{k-t} \mathbb{E}_{s^* \sim \mathcal{L}(s \rightarrow \cdot, k-t, \pi), a \sim \pi(\cdot | s^*), s' \sim \mathcal{P}(\cdot | a, s^*)} [\mathcal{R}(a, s^*, s')] && \begin{array}{l} \text{Expand subscript to} \\ \text{show real distributions.} \end{array} \\
&= \sum_{k=t}^T \gamma^{k-t} \mathbb{E}_{s^* \sim \mathcal{L}(s \rightarrow \cdot, k-t, \pi)} [\mathbb{E}_{a \sim \pi(\cdot | s^*)} [\mathbb{E}_{s' \sim \mathcal{P}(\cdot | a, s^*)} [\mathcal{R}(a, s^*, s')]]] && \begin{array}{l} \text{Expand reward} \\ \text{function definition.} \end{array} \\
&= \sum_{k=t}^T \gamma^{k-t} \left[\sum_{s^* \in S} \mathcal{L}(s \rightarrow s^*, k-t, \pi) \sum_{a \in A} [\pi(a | s^*) \sum_{s' \in S} [\mathcal{P}(s' | a, s^*) \mathcal{R}(a, s^*, s')]] \right] && \begin{array}{l} \text{Show as 3 compound} \\ \text{expectations} \end{array} \\
&= \sum_{k=t}^T \gamma^{k-t} \left[\sum_{s^* \in S} \mathcal{L}(s \rightarrow s^*, k-t, \pi) \sum_{a \in A} [\pi(a | s^*) \sum_{s' \in S} [\mathcal{P}(s' | a, s^*) \mathcal{R}(a, s^*, s')]] \right] && \begin{array}{l} \text{Expand each expecta-} \\ \text{tion into summations.} \end{array}
\end{aligned}$$

What is this actually saying? We need to consider what it means to follow the policy π beginning at state s . We must consider all the actions a_t we could take, and then all the states s_{t+1} those actions could end up in, and then how much reward r_t that would give us. Then, for all those possible states s_{t+1} we could've ended up in, we need to consider each action a_{t+1} we could take, resulting in each state s_{t+2} , and how much reward that would give us. We must continue like this to infinity (or until the episode ends).

3.3 The Bellman Equation

The prior definition of the value function is seldom used in practice because it's very confusing. There is a more concise version, called the Bellman equation, which defines $V(s)$ in terms of $V(s')$:

$$\begin{aligned}
V^\pi(s) &= \mathbb{E}_\pi \left[\sum_{k=t}^T \gamma^{k-t} r_k \mid S_t = s \right] && \text{Original definition} \\
&= \mathbb{E}_\pi [\gamma^0 r_t + \gamma \sum_{k=t+1}^T \gamma^{k-t-1} r_k \mid S_t = s] && \begin{array}{l} \text{Pull out first term of} \\ \text{sum and factor } \gamma \end{array} \\
&= \sum_{a \in A} [\pi(a | s) \sum_{s' \in S} [\mathcal{P}(s' | a, s) (\mathcal{R}(a, s, s') + \gamma \mathbb{E}_\pi [\sum_{k=t+1}^T \gamma^{k-t-1} r_k \mid S_{t+1} = s'])]] && \begin{array}{l} \text{Pull first term out of ex-} \\ \text{pectation} \end{array} \\
&= \sum_{a \in A} [\pi(a | s) \sum_{s' \in S} [\mathcal{P}(s' | a, s) (\mathcal{R}(a, s, s') + \gamma V^\pi(s'))]] && \text{Substitute } V^\pi(s')
\end{aligned}$$

Step 3 is probably the point of most confusion in this. To pull the first term out of the expectation, we need the reward r_t , which requires knowing a_t , s_t , and s_{t+1} . We know s_t , because it is just s , but a_t is determined by the policy and s_{t+1} is determined by the environment dynamics, so we must consider the probability of all actions and next states under the MDP.

4 The Action-Value Function

The action-value function is defined as:

$$Q^\pi(s, a) = \mathbb{E}_\pi [G_t \mid S_t = s, A_t = a]$$

where a can be *any* action, not just an action sampled from π . This is very similar to the value function, but instead of only representing the expected return when following policy π and starting with $S_t = s$, we also condition on starting with taking action $A_t = a$. This means that the Q function tells us how good it is to be in state s , immediately take action a , and then continue to take actions according to the policy π .

4.1 Bellman Equations for the Action-Value Function

4.1.1 V in terms of Q

The only difference between V and Q is how the first action is chosen. In the case of the value function, we always sample the action from the policy at every timestep. For the action-value function we sample the action

from the policy at every timestep *except* the first action. As such, we can write the value function as:

$$V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)}[Q^\pi(s, a)] = \sum_{a \in A} \pi(a|s) Q^\pi(s, a)$$

Because the expectation of the Q function is the value function, it is an unbiased estimator of the value function. This can be very helpful in estimating the value function.

4.1.2 Q in terms of V

We can also write the Q function in terms of the value function as follows:

$$Q^\pi(s, a) = \mathbb{E}_{s' \sim \mathcal{P}(\cdot|a, s)}[\mathcal{R}(a, s, s') + \gamma V^\pi(s')] = \sum_{s' \in S} \mathcal{P}(s'|a, s) [\mathcal{R}(a, s, s') + \gamma V^\pi(s')]$$

4.1.3 Q in terms of next Q

Similarly to the value function, we can also write the action-value function in terms of itself.

$$\begin{aligned} Q^\pi(s, a) &= \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] && \text{Starting definition} \\ &= \mathbb{E}_\pi[r_t + \gamma G_{t+1} \mid S_t = s, A_t = a] && \text{Pull out first term of re-} \\ &= \sum_{s' \in S} [\mathcal{P}(s'|a, s) [\mathcal{R}(a, s, s') + \gamma \sum_{a' \in A} [\pi(a'|s') \mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s', A_{t+1} = a']]]] && \text{Pull first term out of ex-} \\ &= \sum_{s' \in S} [\mathcal{P}(s'|a, s) [\mathcal{R}(a, s, s') + \gamma \sum_{a' \in A} [\pi(a'|s') Q^\pi(s', a')]]] && \text{Substitute } Q^\pi(s', a') \end{aligned}$$

4.2 Formal Action-Value Function Expansion

Similarly to the value function, we can increase the verbosity of the action-value function and fully expand it.

$$\begin{aligned} Q^\pi(s, a) &= \sum_{s' \in S} \mathcal{P}(s'|a, s) [\mathcal{R}(a, s, s') + \gamma V^\pi(s')] && \text{Begin in terms of } V \\ &= \sum_{s' \in S} \mathcal{P}(s'|a, s) \left[\mathcal{R}(a, s, s') + \gamma \sum_{k=t+1}^T \gamma^{k-t-1} \right. && \text{Substitute expanded} \\ &\quad \cdot \left. \left[\sum_{s^* \in S} \mathcal{L}(s' \rightarrow s^*, k-t-1, \pi) \sum_{a^* \in A} [\pi(a^*|s^*) \sum_{s'' \in S} [\mathcal{P}(s''|a^*, s^*) \mathcal{R}(a^*, s^*, s'')]] \right] \right] && \text{version of value func-} \\ &\quad \cdot \left. \left[\sum_{s^* \in S} \mathcal{L}(s' \rightarrow s^*, k-t-1, \pi) \sum_{a^* \in A} [\pi(a^*|s^*) \sum_{s'' \in S} [\mathcal{P}(s''|a^*, s^*) \mathcal{R}(a^*, s^*, s'')]] \right] \right] && \text{tion.} \end{aligned}$$

5 The Advantage Function

Because the Q function does not force us to only consider actions sampled from π , it can be quite useful in determining if an action which would not be sampled in our current policy is better or worse than the action that would be sampled from our current policy. We can quantify this with the advantage function, defined as follows:

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$$

This tells us how much better it is to take action a , and then follow π , than it is to follow only actions sampled from π .

References

- [1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA: MIT Press, 2018. [Online]. Available: <https://www.andrew.cmu.edu/course/10-703/textbook/BartoSutton.pdf>. [Accessed: Jun-2026].
- [2] D. Silver, “Introduction to Reinforcement Learning,” University College London and DeepMind, 2015. [Online]. Available: <https://davidstarsilver.wordpress.com/teaching/>. [Accessed: Jun-2026].